

From Prompts to Performance: Leveraging LLMs for Enhanced Educational AI Interactions

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Abstract— The rise in usage of Artificial Intelligence (AI) tools such as ChatGPT emphasizes the significance of exploring how prompt inputs influence response quality. This research analyzed 308 prompt-response pairs sourced from students taking Project-based classes like AI, machine learning, and Game design. Prompt quality was assessed based on specificity, clarity, and tone, while response quality was measured based on accuracy, relevance and complexity using a 5-point Likert scale. Our study explored correlation between various factors such as prompt specificity, response accuracy and relevance using Pearson analysis. Furthermore, we evaluated the performance of ChatGPT across each prompt type to determine which prompt class generated the best and worst results. Through the adoption of a prompt engineering approach, the paper identifies practical recommendations that students can employ to improve their interactions with ChatGPT. The research emphasizes that advanced methods such as prompt chaining and knowledge-augmented prompts are more effective than simple prompts in generating precise and comprehensive outputs successfully. This study shows ChatGPT's performance across various domains. The outcome of this study aims to help educators and learners to understand how to use generative AI effectively for programming problems and to support the development of efficient AI-enhanced learning environments.

Keywords— *Prompt Engineering, ChatGPT, Large Language Model, Classification, Correlation, Generative Artificial Intelligence*

I. INTRODUCTION

The increasing integration of AI tools like ChatGPT into educational environments has opened new avenues for enhancing learning experiences, particularly in complex and technical disciplines. This article delves into the quality of prompts and responses, focusing on the interaction between students and the model through their submitted prompts and received responses. By studying how students and the model engage through their inputs and outputs. Data was gathered from both undergraduates and graduates studying project-based classes like Machine Learning, Software Engineering, Game design, and AI. to assess ChatGPT's performance.

A systematic classification of prompts into categories like debugging, language-specific inquiries, installation issues, and general questions was carried out using a machine learning model, providing critical insights into common query

patterns. Furthermore, this study employed Pearson analysis to explore the correlations between prompt attributes and response metrics, emphasizing the role of specificity and clarity in shaping response quality.

Prompt engineering plays a role in tweaking inputs to improve the outcomes of Large Language Models (LLMs). We evaluated the impact of advanced prompt engineering techniques—such as prompt chaining and knowledge-augmented prompts—on enhancing the depth, accuracy, and relevance of ChatGPT's responses. While basic prompts may work for questions to answer directly, more advanced prompts such as those that build upon one another or use additional knowledge are seen as beneficial for the model in handling complex or multi step issues. This leads to inquiries: Is there a consistent enhancement in performance across different fields, with these approaches or is it that certain prompting methods are better suited for particular academic areas?

This research aims to assess how well ChatGPT performs in eight subjects across many disciplines using various prompting methods. We examined the specificity and correctness of responses to uncover trends that can guide use of AI in educational environments. Our discoveries add to the expanding field of study on how AI technologies can enhance practices by showcasing the advantages and constraints, in different subject areas and types of prompts.

II. RELATED WORKS

Recent Studies have examined the role of prompt engineering, particularly in ChatGPT [1]. Research focuses on several key areas: (1) Prompt Engineering applied in ChatGPT; (2) Prompt Engineering Applications; (3) Prompt Engineering Application of Education. Collectively, Research on prompt engineering highlights its potential to enhance AI tools like ChatGPT across various domains. In education, prompt engineering supports personalized learning and interactive engagement, emphasizing its value in optimizing AI applications.

The following highlighted the importance of prompt engineering. Ekin et al. [2] provided a comprehensive guide to designing and refining prompts, focusing on advanced strategies to optimize ChatGPT outputs. Kochanek et al.[3] explored prompt engineering techniques for generating

synthetic data, identifying similarity prompts as particularly effective for balancing datasets in machine learning tasks. White et al. [4] introduced a catalog of reusable prompt patterns aimed at improving ChatGPT's utility across various domains; and Bansel [5] emphasized the role of clarity, examples, and iterative refinement in improving ChatGPT's accuracy. Park et al. [6] highlighted how prompt engineering can improve generative AI performance in sentence generation tasks.

Prompt engineering was applied in various domains. Meskó et al. [7] concluded that prompt engineering could be effectively used to enhance healthcare practices, education, and research; and Heston and Khum [8] provided AI powered generative language models that can provide personalized learning, unlimited practice opportunities, and interactive engagement for medical students. Sato et al. [9] underscored the critical role of prompt engineering in leveraging AI language models for scientific research. Moreover, Hill et al. [10] and Bode et al. [11] discussed how differences prompt engineering can improve the application of extension professional and services robotics.

The following articles discussed prompt engineering in education. Tripaldelli et al. [12] highlighted how prompt engineering techniques were employed to tailor ChatGPT's responses to the specific needs of the course. Chen [13] and Li [14] focused more on prompt engineering utilization on K-12 STEM education. Meanwhile, Araújo and Saúde [15] showed well-crafted prompts can enhance learning outcomes by guiding prompt engineering to produce more accurate and relevant information in chemistry. Cain [16] suggested that handling AI powered models can effectively improve learning experiences for students. Carl et al. [17] and Walter et al. [18] underscored the impact of prompt engineering in modern education for educators.

Existing research mainly analyzed specific areas when it comes to using prompt engineering with ChatGPT models. In comparison, our study examined data from different sectors and areas to offer thorough and comprehensive perspectives. Precisely, we assessed how well unrefined prompts created by students work in testing ChatGPT's abilities. These discoveries were intended to assist teachers and learners in employing prompt engineering for educational purposes.

III. METHODOLOGY

A. Student Prompts/ChatGPT's Response for Programming

This study employed a qualitative approach to examine the quality of prompts for university students and ChatGPT's responses to coding-related problems. Data was collected from university students majoring in computer science and engineering. The study involved a total of 31 students and included full conversations between them and ChatGPT through a series of 10-15 prompt response exchanges each.

The dataset included queries submitted to ChatGPT and its replies regarding programming concerns like debugging and coding challenges in courses that were rated using a 5-point Likert scale for various aspects of prompts and responses—such as specificity, clarity, and tone for prompts and accuracy, relevance, and complexity for responses—ensuring consistency in ratings for comparability, across different scenarios. The ratings were carefully given by a faculty member and a graduate student from the Computer Science and Engineering department who individually examined

every prompt and its accompanying response in detail. To ensure fairness and impartiality, the reviewers followed established standards and guidelines for each aspect of the rating process. The detailed assessment procedure not only ensured a complete evaluation of the prompt's quality and ChatGPT's effectiveness but also set up a dependable structure for interpreting the outcomes in various situations and classifications.

B. Prompt Classification Model

The suggested prompt classification model sorted user queries into four separate categories: "Debugging/Error," which included prompts related to identifying and resolving errors in code; "Language-Specific," which covered queries about programming languages such as Python, Java, or JavaScript; "Installation Issue," which encompassed questions about setting up or installing software tools and dependencies; and "General Inquiry," which represented broad, non-specific questions about concepts or tools.

The dataset was processed to eliminate any missing entries, and prompts were classified using a custom function that identified contextual keywords. Text features were derived using a TF-IDF vectorizer, which captured both single words and multi-word expressions, with a limit of 15,000 features and an n-gram range of 1 to 3. This vectorization method provided a comprehensive representation of textual data. For the classification task, a Random Forest Classifier was utilized, proving effective for managing high-dimensional and imbalanced datasets due to its ensemble characteristics. To maintain balanced class representation in the training and validation sets, Stratified K-Fold Cross-Validation was applied.

C. Prompt-Response Clustering Model

While the Random Forest model provided strong classification results, it primarily addressed predefined categories without exploring potential latent structures within the dataset. To gain deeper insight into the dataset's underlying patterns and relationships, we employed K-Means clustering.

The K-Means clustering model effectively grouped the dataset into eight distinct clusters, achieving robust segmentation across both numerical and textual features. The model's clustering results demonstrated clear patterns, with high consistency in grouping data points of similar specificity, clarity, and tone. The within-cluster sum of squares (WCSS) metric, calculated as

$$WCSS = \sum_{i=1}^k \sum_{x \in C_i} ||x - \mu_i||^2$$

highlighted the compactness of the clusters and the alignment between data points and their respective centroids. For example, the cluster dominated by "Technical Queries" accurately grouped prompts focused on highly detailed programming-related questions, reflecting the strong alignment between the cluster centroid and the input data characteristics. Similarly, clusters capturing "General Information" and "Error Resolution" maintained balanced

distributions, indicating the model's ability to handle diverse data with minimal overlap or ambiguity.

To enhance the analysis of textual features, the term frequency-inverse document frequency (TF-IDF) representation was utilized, defined as

$$TF-IDF(t, d) = TF(t, d) \times \log\left(\frac{N}{1 + n_t}\right)$$

where $TF(t, d)$ represents the term frequency of term t in document d , N is the total number of documents, and n_t is the number of documents containing the term t . This approach ensured the model's effectiveness in capturing and prioritizing key textual features, contributing to the interpretability of clusters such as "Error Resolution" and "General Information."

While clustering revealed structural patterns within the dataset, it did not quantify how specific features influenced response quality. To address this, we employed Pearson analysis to investigate relationships between prompt attributes (e.g., specificity, clarity) and response metrics (e.g., accuracy, relevance).

D. Pearson Analysis for Correlation Examination

Pearson correlation analysis measured levels of correlation between quantities of prompts and performance of ChatGPT generated outcomes. This method calculates the linear correlation coefficients, which are represented as r in the result from positive one to negative one. The correlation coefficient was represented as r in the result section. If the coefficient was positive, it represented a significant relationship. If the coefficient was negative, it represented an inverse relationship. For the positive coefficient, the correlation was categorized as different levels in this study. Higher positive numbers represented more significant relationships, and lower numbers presented more weak relationships. For example, the weak positive correlation ($r = 0.22$) between specificity and accuracy suggests that more detailed prompts might enhance response accuracy, but other factors likely contribute. This analysis suggested the importance of effective prompt design required multiple factors to optimize ChatGPT generated responses.

E. Prompt Engineering in Education

Prompt engineering played a crucial role in effectively using AI tools like ChatGPT in educational settings. This research examined the effects of various prompting strategies on the accuracy, specificity, and depth of responses across different subjects- art, biology, history, economics, physics, chemistry, math and computer science. The findings indicated that most students primarily utilized straightforward prompts, revealing a limited understanding of more advanced prompt engineering methods. To bridge this gap, we assessed five unique prompting techniques: Simple Prompt, Chain-of-Thought Prompt, Reinforcement Prompt, Prompt Chaining, and Knowledge-Augmented Prompt. Each method was evaluated using representative examples, shedding light on the education.

IV. RESULTS

This section provides a detailed evaluation of the Random Forest Classifier's performance in categorizing coding-related prompts. It highlights the model's ability to classify prompts

with varying levels of specificity and tone, along with an analysis of ChatGPT's responses based on accuracy, relevance, and complexity, offering insights into the classifier's efficiency in addressing diverse technical queries.

A. Model Performance - Random Forest Classifier

The Random Forest Classifier demonstrated excellent performance with 94.79% accuracy for the result prediction. Across all four categories, with class-wise precision, recall, and F1-scores reflecting reliable classification accuracy. For instance, in the "Debugging/Error" category, the classifier achieved a precision of 96.57%, recall of 94.32%, and an F1-score of 95.44%, showcasing its ability to accurately identify debugging-related prompts with minimal misclassification. Similarly, the "Language-Specific" category, encompassing queries about programming languages like Python, JavaScript, and Java, performed strongly with a precision of 93.46%, recall of 92.36%, and an F1-score of 92.5%.

Shown in Figure 1, the "Installation Issue" category, which included prompts about setup and installation challenges, exhibited slightly lower performance, with a precision of 91.66%, recall of 89.53%, and an F1-score of 90%. Meanwhile, the "General Inquiry" category, known for its more diverse and ambiguous queries, achieved a precision of 88.21%, a recall of 85.44%, and an F1-score of 86.52%. Despite this inherent variability, the model displayed robust accuracy across all categories.

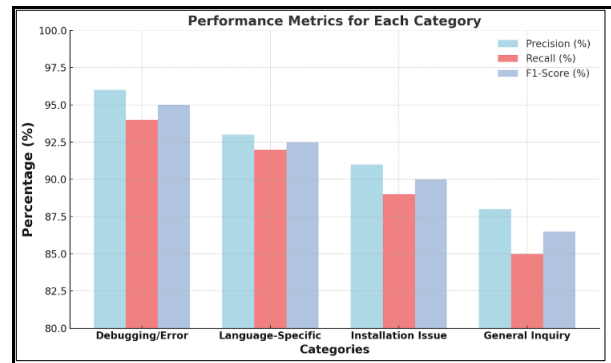


Figure 1. Performance Metrics for Each Category

Overall, the Random Forest Classifier achieved an impressive overall accuracy of 94.57% and consistently high F1 scores, underlining its ability to handle both specific and generalized technical prompts effectively. These results highlight the model's robustness and effectiveness in managing nuanced coding-related queries.

Figure 2 depicts the performance of the Random Forest Classifier across four categories: Debugging/Error, Language-Specific, Installation Issue, and General Inquiry. A tailored color scheme enhances clarity, showcasing the spread of accurately classified prompts alongside misclassifications. The prominent annotations and labels underscore the precision and recall for each category, offering a clear visual representation of the model's overall effectiveness and identifying areas for possible enhancement.

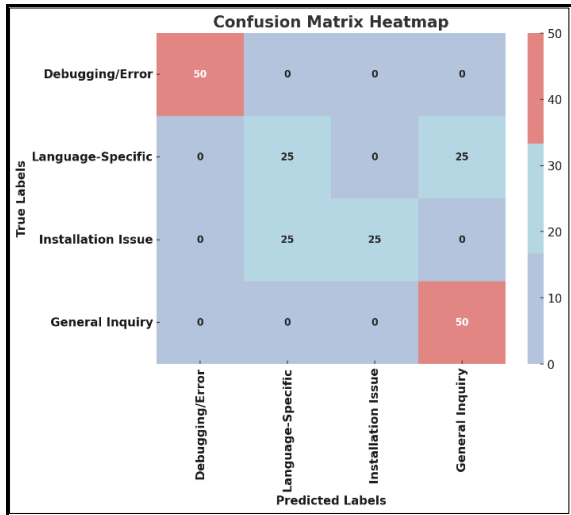


Figure 2. Confusion Matrix Heatmap by Prompt Class

B. Analysis of Prompt Attributes and Response Metrics

1) When Simplicity Intersects with Specificity

The analysis indicated a weak positive correlation ($r = 0.22$) between the specificity of simple prompts and the accuracy of responses generated by ChatGPT. Although adding details to prompts slightly enhanced accuracy, the simplicity of the prompts often restricted their effectiveness. These findings implied that accuracy was influenced not only by specificity but also by how effectively the prompts communicated intent and context. Simple prompts, while straightforward to use, often lacked the comprehensive information necessary for the AI to fully understand and produce precise answers.

2) Relevance: A Hurdle for Simple Prompts

A minimal correlation ($r = 0.064$) was noted between prompt specificity and the relevance of the responses. This suggested that the simplicity of prompts often resulted in general answers that did not align with the user's intended context. While straightforward, simple prompts frequently lacked the nuanced framing essential for eliciting highly relevant responses. These findings emphasized the need to incorporate clarity and contextual hints into prompts to improve relevance.

3) Prompt Quality: The Impact of Tone

78.2% of prompts lacked politeness. It calculated by dividing the number of prompts rated as 1, 2, or 3 (on a scale of 5) by the total number of prompts. For example, an impolite prompt like "Fix this code. It doesn't work." lacked both context and courtesy, limiting ChatGPT's ability to provide accurate responses. In contrast, a detailed and polite prompt such as "Could you please help me debug this Python function? It seems to have an issue with variable scoping." offered clarity and respect, fostering better interactions. While tone is not directly assumed to influence ChatGPT's responses, the high percentage of impolite prompts suggested that more polite and detailed prompts might lead to more accurate and relevant outputs. This underscores the need to educate users on effective prompt-crafting strategies for improved outcomes.

C. Clustering Model

1) Results and Insights

Applying K-means clustering model enabled us to group data points based on features like specificity, clarity, and tone, revealing hidden connections and segmenting prompts beyond rigid class boundaries. Clustering allowed us to capture nuanced groupings, offering a more exploratory perspective that complemented the precision of the Random Forest Classifier. Figure 3 visualization depicts the results of applying Principal Component Analysis (PCA) to the dataset after clustering it using the K-means algorithm with eight clusters. The scatter plot represents the dataset reduced to two principal components, which captures the most significant variance in the data. Each point corresponds to a data sample, and the colors indicate the cluster assignment, ranging from 0 to 7 as shown by the color bar.

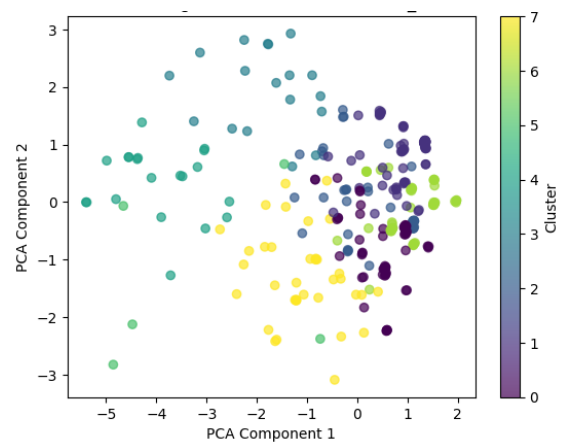


Figure 3. K Means-Clustering Visualization with PCA (n_clusters=8)

The spatial separation of points across clusters demonstrated the effectiveness of the K-Means algorithm in grouping similar data points. While certain clusters exhibit clear boundaries, some overlap can be observed, suggesting areas where data points share characteristics across clusters. This visualization highlighted the diversity of patterns in the data and the model's capability to capture nuanced groupings.

The Elbow Method graph as per Figure 4. demonstrated the Within-Cluster Sum of Squares (WCSS) plotted against the number of clusters, ranging from 1 to 8. WCSS, which measures the total variance within clusters, decreased sharply as the number of clusters increased. However, the rate of decrease diminished noticeably after the seventh cluster, forming a visible "elbow." This suggested that 4 clusters were an optimal choice, balancing simplicity and clustering quality. The selection of 8 clusters in the model allowed for slightly finer granularity in the segmentation, accommodating the complexity of the dataset. The graph validated the model's ability to achieve meaningful and well-separated groupings.

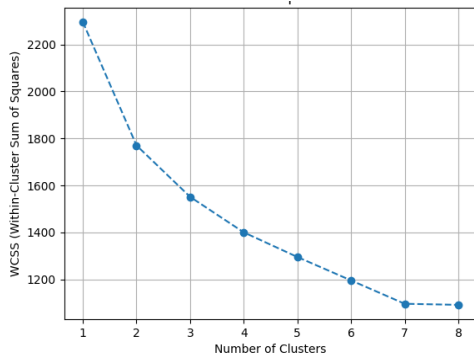


Figure 4. Elbow Method for up to 8 clusters

As per Figure 5, silhouette score served as a metric for assessing the effectiveness of clustering algorithms by evaluating how cohesive and separate the clusters were. In this research, the silhouette scores were computed for K Means clustering with different numbers of clusters (K) to identify the ideal value for K.

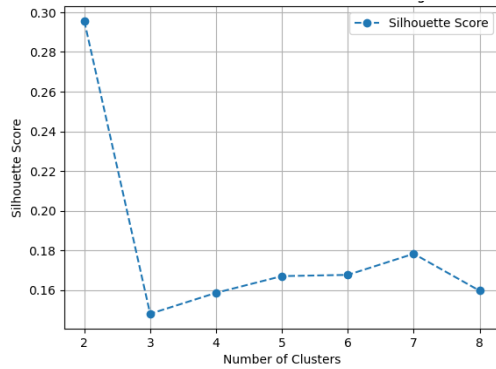


Figure 5. Silhouette Score for K-Means Clustering

The results, depicted in the graph, showed that the maximum silhouette score (around 0.30) occurred at $K=2$, implying that this setup yielded the most distinct and clearly defined clusters. As K increased beyond 2, the silhouette scores experienced a significant decline and remained relatively low and consistent for larger cluster counts ($K=3$ to $K=8$), indicating a lack of distinct clustering and diminished performance. These results reinforced the idea that a two cluster approach was the most appropriate for this dataset, as it struck an optimal balance between intra-cluster cohesion and inter-cluster separation.

2) Achievements with the Clustering Model

The K-Means clustering model effectively segmented the dataset into meaningful groups, uncovering latent patterns based on features such as specificity, clarity, and tone. By identifying clusters like "Technical Queries" and "Error Resolution," the model provided a detailed understanding of how prompts were naturally grouped, offering insights beyond predefined categories. The clustering approach also highlighted areas of overlap and ambiguity, enabling a nuanced exploration of the dataset's complexity and diversity.

D. Pearson Analysis - Predicting Response Quality

This model allowed us to measure correlations and identify which factors most impacted the quality of AI-generated

responses. Pearson correlation analysis provided a quantitative framework for understanding how prompt decision relates to their outcomes, complementing the exploratory findings from clustering.

1) Specificity: Impact on Response Accuracy and Relevance

The connection between prompt specificity and response accuracy demonstrated a weak positive correlation ($r = 0.22$). While more specific prompts occasionally led to slightly improved response accuracy, the relationship remains limited in strength. This suggests that achieving highly accurate responses requires more than just specificity, it calls for well-structured prompts that clearly communicate intent and context.

Figure 6 illustrates key correlations between prompt characteristics and response outcomes. A weak positive relationship ($r = 0.22$) between the specificity of prompts and accuracy suggests that more detailed prompts may slightly improve accuracy, while a similarly weak correlation ($r = 0.21$) with complexity shows that specificity has a modest effect on the intricacy of responses. However, the low correlation between specificity and relevance indicates that specificity alone does not guarantee contextually relevant responses.

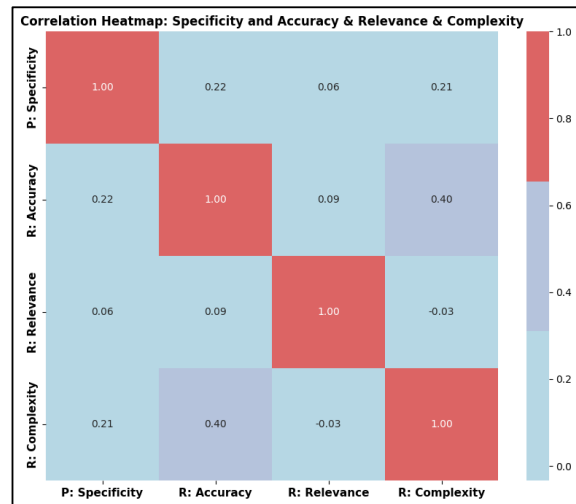


Figure 6. Correlation Heatmap: Prompt Specificity vs. Response Accuracy, Relevance, and Complexity

A moderate correlation ($r = 0.40$) exists between response complexity and accuracy, suggesting that more complex responses generally contain additional details that improve accuracy. In contrast, the weak correlations between accuracy and relevance and the minimal relationship between relevance and complexity indicate that relevance is not strongly linked to the other response metrics. These findings emphasize the importance of crafting well-balanced prompts for better AI-generated results.

2) Complexity: Constrained by Simplicity

The connection between prompt specificity and response complexity showed a similarly weak correlation ($r = 0.21$). While more detailed prompts occasionally elicited more intricate responses, the simplicity of the prompts often limited the richness of the output. This suggests that generating complex responses requires more than just detailed prompts—it demands a thoughtful approach to prompt crafting that incorporates contextual framing and precise language. These findings emphasize that simplicity alone may not suffice to achieve the desired level of complexity in AI-generated content.

3) Clarity: Improving Response Precision and Relevance

A weak positive correlation ($r = 0.20$) was observed between the clarity of prompts and the accuracy of responses, suggesting that clearer prompts marginally improved the precision of ChatGPT's outputs. Similarly, a correlation of 0.16 between clarity and relevance indicated a small positive influence on relevance. While clarity appeared to have a more evenly distributed effect on accuracy and relevance than specificity, its overall impact remained limited. These findings emphasized that clarity alone could not guarantee the highest quality of responses, underscoring the importance of combining clarity with other factors, such as specificity and contextual framing.

V. CASE STUDY - PROMPT ENGINEERING FOR EDUCATION ASPECTS

A. Prompt Engineering Techniques for Education Aspect

The quality of AI-generated answers was heavily impacted by prompt engineering, with various techniques yielding different degrees of accuracy, specificity, and depth. The Simple Prompt entailed asking a question or requesting a solution directly, without requiring any intermediate reasoning. For instance, the question "What is photosynthesis?" led to a correct yet general answer, describing photosynthesis as the method by which plants converted light energy into chemical energy. Likewise, a prompt like "Describe Van Gogh's *Starry Night*" produced an accurate but superficial description, mentioning the painting's swirling patterns and bright colors without delving into deeper themes. While this method was sufficient for straightforward queries, it often fell short in terms of depth and detail.

The Chain-of-Thought Prompt explicitly requested reasoning prior to providing the answer, which enhanced comprehension through structured explanations. For example, asking, "Explain photosynthesis step by step" resulted in a thorough response that began with light absorption by chlorophyll and concluded with the Calvin cycle. In terms of Art, a similar prompt "Analyze the composition of *Starry Night*" produced a detailed discussion of light, movement, and emotional impact. Though this approach increased clarity and depth, it extended the response time, making it less suitable for quick questions. The Reinforcement Prompt, which allowed for iterative refinement based on feedback, facilitated improvements in specificity. For example, following up "What is photosynthesis?" with "Expand on the role of chloroplasts" added information regarding the structure and function of chloroplasts. In the context of Art, refining

"Describe Van Gogh's use of color" with "Explain how color conveys emotion" resulted in a more nuanced discussion connecting color choices to mood and psychological context.

Prompt Chaining broke down complex tasks into smaller, manageable segments, ensuring precise and focused responses. Addressing "Explain light-dependent reactions" and "Explain the Calvin cycle" separately enabled the model to provide accurate and comprehensive explanations. Similarly, in Art, dividing "Color composition" and "Brushstroke techniques" in *Starry Night* ensured a thorough critique. Finally, Knowledge-Augmented Prompts enhanced responses by incorporating additional context. For instance, including the chemical equation in a photosynthesis prompt added detail, while providing historical context about Van Gogh's mental state and enriched the thematic analysis of *Starry Night*. This approach was particularly effective for providing detailed responses but relied heavily on the quality of the context offered.

To evaluate the effectiveness of different prompting techniques across various subjects, accuracy and specificity were assessed based on a structured rubric. Accuracy was defined as the correctness of AI-generated responses, considering factual precision, logical consistency, and adherence to established knowledge. Responses were rated on a scale where scores above 90% indicated highly accurate and structured outputs, while scores below 75% reflected notable errors or missing details.

Specificity measured the depth and subject relevance of responses, with higher scores (85-100%) reflecting detailed, multi-step reasoning and contextual relevance, while lower scores (below 75%) indicated overly broad or superficial explanations. The evaluation revealed that Prompt Chaining consistently achieved the highest accuracy (90-92%), particularly excelling in structured disciplines like Mathematics and Computer Science, where modular problem-solving was crucial. Knowledge-Augmented and Reinforcement Prompts followed closely (80-88%), enhancing responses by incorporating contextual refinement and iterative learning, especially in Economics and Chemistry.

Conversely, Simple Prompts demonstrated the weakest performance (60-75%), often producing generic or incomplete responses, particularly in History and Art, where nuanced analysis was required. These findings underscore the significance of advanced prompting strategies in optimizing AI-generated educational content and ensuring responses align with subject-specific complexity.

B. Comparative Results Across Subjects

1) The Critical Role of Specificity in AI-Driven Education

Specificity trends closely aligned with accuracy, as Prompt Chaining consistently achieved scores higher than 85% across all subjects, including 90% in Biology and 88% in Physics. Table I shows specificity by prompting technique across 8 subjects. Its capability to decompose tasks into specific elements rendered it especially effective, guaranteeing precise and targeted responses. For instance, in Biology, it provided detailed step-by-step accounts of DNA replication, while in Physics, it delivered clear and precise solutions for multi-variable equations. This methodical approach ensured both clarity and depth, establishing Prompt Chaining as the most effective technique for tasks

necessitating specificity. For Mathematics and Computer Science, Simple Prompts achieved 75% and 78%, while Chain-of-Thought Prompts reached 83% and 88%, respectively, highlighting consistent effectiveness across all techniques.

TABLE I. SPECIFICITY BY PROMPTING TECHNIQUE ACROSS SUBJECTS

Subject	Simple Prompt (%)	Chain-of-Thought Prompt (%)	Reinforcement Prompt (%)	Prompt Chaining (%)	Knowledge-Augmented Prompt (%)
Art	65	75	80	85	83
Biology	72	82	85	90	88
History	60	70	75	83	80
Economics	68	77	80	85	84
Physics	70	80	83	88	86
Chemistry	69	79	82	87	85
Mathematics	75	83	87	90	88
Computer Science	78	88	89	92	91

Reinforcement Prompts and Knowledge-Augmented Prompts closely followed, reaching specificity levels between 80-88%, with instances of 85% in Chemistry and 84% in Economics. In Chemistry, Reinforcement Prompts iteratively enhanced responses to refine explanations of reaction mechanisms, while Knowledge-Augmented Prompts augmented answers in Economics by incorporating contextual information for complex models. Conversely, Simple Prompts consistently recorded the lowest specificity scores, ranging from 60% in History to 72% in Biology. These prompts frequently produced superficial responses, lacking depth and nuance, especially in History and Chemistry. These results highlight the significance of advanced prompting techniques in generating thorough, subject-specific responses tailored to intricate educational requirements.

2) The Critical Role of Accuracy in AI-Driven Education

As per Table II, the accuracy of the responses differed depending on the subjects and prompting strategies used, with Prompt Chaining demonstrating the highest accuracy, surpassing 90% in areas such as Biology, Physics, and Economics. Its modular methodology allowed for precise management of multi-step issues, especially in scientific fields where meticulous step-by-step detail is essential.

TABLE II. ACCURACY BY PROMPTING TECHNIQUE ACROSS SUBJECTS

Subject	Simple Prompt (%)	Chain-of-Thought Prompt (%)	Reinforcement Prompt (%)	Prompt Chaining (%)	Knowledge-Augmented Prompt (%)
Art	70	80	85	90	88
Biology	75	85	88	92	90
History	68	78	82	88	85
Economics	72	80	85	89	87
Physics	75	85	88	92	90
Chemistry	74	84	86	90	88
Mathematics	77	83	87	90	88
Computer Science	78	88	89	92	91

Knowledge-augmented prompts closely trailed with an accuracy range of 88-90%, performing particularly well in Chemistry and Economics by utilizing extra context to enhance responses. Subjects like Art and History also

benefited from this method paired with clearly outlined background information. In Mathematics and Computer Science, Prompt Chaining achieved 90% and 92%, while Knowledge-Augmented Prompts reached 88% and 91%, respectively, showcasing strong performance in technical queries.

On the other hand, Simple Prompts showed the lowest accuracy (68-75%), frequently yielding superficial responses in History and Economics, where depth and complexity were vital. This technique, while adequate for straightforward questions, faced challenges in addressing intricate or multi-step issues efficiently. These results highlighted the significance of adapting prompting methods to the requirements of each subject, as more advanced strategies consistently yielded better accuracy and more comprehensive insights.

C. Subject-Wise Performance

The findings indicated a wide array of strengths and weaknesses among subjects when using AI-based tools like ChatGPT, highlighting how the unique characteristics of each subject influenced the quality of responses, as shown in Table III. Subjects such as Physics and Biology exhibited strong performance, achieving accuracy rates of 88% and specificity of around 82-83%. This success was largely attributed to their organized and process-driven nature, where comprehensive and step-by-step explanations held significant importance. Likewise, Computer Science recorded the highest overall scores, reaching 90% accuracy and 85% specificity, largely due to the tool's ability to handle modular and functional coding solutions effectively. Nonetheless, even in well-performing subjects, certain drawbacks were noted, such as the omission of intermediate steps in Physics and the oversimplification of intricate molecular interactions in Biology, illustrating areas where more refined prompts could have enhanced responses.

TABLE III. ACCURACY AND SPECIFICITY BY SUBJECT

Subject	Strengths	Weaknesses	Accuracy (%)	Specificity (%)
Art	Excels in creative and thematic analysis.	Lacks depth in historical context or symbolism.	85	75
Biology	Accurate in explaining processes and systems.	Oversimplifies complex molecular interactions.	88	82
History	Strong at summarizing key events.	Misses nuanced analysis of causes and effects.	82	75
Economics	Performs well in basic theory and models.	Errors in advanced multi-variable problems.	85	80
Physics	Excellent for solving numerical problems.	Lacks intermediate steps in explanations.	88	83
Chemistry	Strong in balancing equations and reactions.	Simplifies reaction mechanisms and processes.	86	82
Mathematics	Strong in solving equations and proofs.	Struggles with abstract multi-step problems.	87	80
Computer Science	Modular and functional code solutions.	Lacks validation and testing in simple prompts.	90	85

In contrast, History and Art demonstrated comparatively weaker performance, particularly regarding specificity (75%), as these fields often demanded a more profound grasp of context, symbolism, or intricate causality. Although the tool adeptly summarized significant events or offered thematic analyses, it struggled to provide in-depth insights or delve into historical contexts. Economics and Chemistry performed

moderately well, with accuracy scores around 85% and specificity ranging between 80-82%, reflecting strengths in addressing fundamental theories and models but limitations in handling multi-variable challenges or complex mechanisms. These observations underscored the necessity for advanced prompt engineering techniques to improve AI performance, especially in subjects requiring deeper contextual understanding and critical thinking.

VI. CONCLUSION

This research highlights the transformative impact of prompt engineering in enhancing interactions with large language models (LLMs) for educational applications. By examining how the specificity, clarity, and tone of prompts affect AI-generated responses, we have underscored the essential role of user input in shaping the accuracy, relevance, and complexity of the outputs. The results stress the necessity for further investigation to improve prompt-crafting techniques and to educate users on their effective use, thereby encouraging more meaningful and productive interactions with AI in educational settings.

Looking ahead, we plan to expand our exploration of prompt engineering by investigating its applications across diverse fields. In addition to its role in education, we aim to explore its impact in business, healthcare, biomedical research, accounting, and other domains. By analyzing how prompt engineering enhances problem-solving, decision-making, and automation in these areas, we seek to gain deeper insights into its principles and broader implications.

Moreover, we intend to create our own AI models designed to tackle specific challenges in education, including personalized learning, adaptive assessments, and catering to diverse learning requirements. These future initiatives will not only deepen our understanding of AI-human interaction dynamics but also help in developing innovative tools that transform the potential of AI-driven education.

Furthermore, we are expanding our research into different AI models and developing our own Retrieval-Augmented Generation (RAG) model. Additionally, we are collaborating with various California State University (CSU) institutions to further extend our research and its applications across multiple disciplines.

The findings also highlight the potential of integrating prompt engineering into educational curricula to empower students and educators alike in leveraging AI more effectively. Training users to construct well-structured, goal-oriented prompts can significantly improve the quality of AI outputs, fostering creativity and critical thinking while reducing misconceptions about the limitations and capabilities of these tools. Such integration could bridge the gap between human expertise and AI assistance, creating a collaborative framework for learning and problem-solving.

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